

EXPERIMENT - 12

Aim

To write a menu driven C program using switch case to input a positive integer and:

- (a) find the sum and product of its first and last digit
- (b) count the digits that are lesser than, equal to, and greater than a given digit k

Problem Statement

Prepare the algorithm and test case table. Then write, compile, execute, and test the C program against the prepared use cases.

Write a menu-driven C program using switch-case to input a positive integer and perform one of the following operations based on the user's choice:

- Find the sum and product of the first and last digits. **E.g.** Number: 12345, Sum = 6, Product = 5
- Count how many digits are lesser than, equal to, and greater than a given digit k. **E.g.** Number: 12345, k=3, Count_lesser=2, Count_equal=1, Count_greater=2

Algorithm

1. Start.
2. Display menu:
 1. Sum and product of first and last digit
 2. Count digits lesser, equal, greater than k
 3. Exit
3. Read choice.
4. If choice = 1:
 - a. Read num
 - b. $\text{last} = \text{num} \% 10$
 - c. Keep dividing num by 10 till one digit is left, this is first
 - d. $\text{sum} = \text{first} + \text{last}$, $\text{product} = \text{first} * \text{last}$
 - e. Print sum and product
5. If choice = 2:
 - a. Read num and k
 - b. Take each digit of num ($\text{num} \% 10$) and compare with k
 - c. Increment `count_lesser` / `count_equal` / `count_greater` accordingly
 - d. Print the three counts
6. If choice = 3, exit the program.
7. If choice is invalid, print an error message.
8. Repeat steps 2 to 7 until choice = 3.
9. Stop.

Test Case Table

The following test cases were prepared before running the program, and later checked against the actual output.

#	Menu Option	Input	Expected Output	Actual Output
1	1	num = 12345	Sum = 6, Product = 5	Match
2	2	num = 12345, k = 3	Count_lesser = 2, Count_equal = 1, Count_greater = 2	Match

Program Code

```
#include <stdio.h>

int main() {
    int num, choice, k;

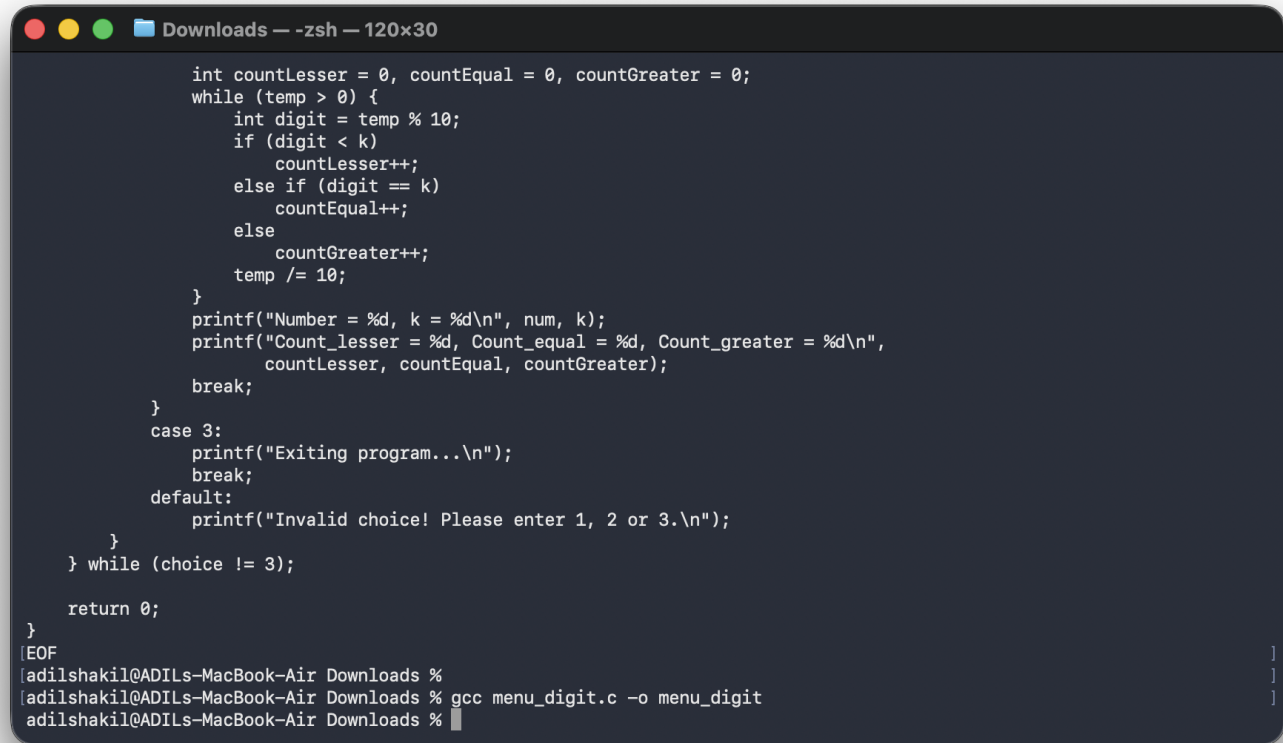
    do {
        printf("\n----- MENU ----- \n");
        printf("1. Sum and Product of first and last digit\n");
        printf("2. Count digits lesser, equal, greater than k\n");
        printf("3. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch (choice) {
            case 1: {
                printf("Enter a positive integer: ");
                scanf("%d", &num);
                if (num <= 0) {
                    printf("Invalid input! Please enter a positive integer.\n");
                    break;
                }
                int temp = num;
                int lastDigit = temp % 10;
                while (temp >= 10)
                    temp /= 10;
                int firstDigit = temp;
                int sum = firstDigit + lastDigit;
                int product = firstDigit * lastDigit;
                printf("Number = %d\n", num);
                printf("First Digit = %d, Last Digit = %d\n", firstDigit, lastDigit);
                printf("Sum = %d, Product = %d\n", sum, product);
                break;
            }
            case 2: {
                printf("Enter a positive integer: ");
                scanf("%d", &num);
                if (num <= 0) {
                    printf("Invalid input! Please enter a positive integer.\n");
                    break;
                }
                printf("Enter digit k (0-9) to compare: ");
                scanf("%d", &k);
                int temp = num;
                int countLesser = 0, countEqual = 0, countGreater = 0;
                while (temp > 0) {
                    int digit = temp % 10;
                    if (digit < k)
                        countLesser++;
                    else if (digit == k)
                        countEqual++;
                    else
                        countGreater++;
                    temp /= 10;
                }
                printf("Number = %d, k = %d\n", num, k);
                printf("Count_lesser = %d, Count_equal = %d, Count_greater = %d\n",
                    countLesser, countEqual, countGreater);
                break;
            }
            case 3:
                printf("Exiting program...\n");
                break;
            default:
                printf("Invalid choice! Please enter 1, 2 or 3.\n");
        }
    } while (choice != 3);

    return 0;
}
```

Output

Screenshots of compiling, running and testing the program are given below.

A terminal window titled "Downloads — zsh — 120x30" displays C code for counting digits in a number. The code includes a while loop to process the number digit by digit, incrementing counters for less than, equal to, and greater than a specified digit 'k'. It also includes a menu system with three options: 1 (run), 2 (test), and 3 (exit). The terminal shows the code being entered, followed by EOF, and then the compilation command 'gcc menu_digit.c -o menu_digit' being executed. The prompt returns to the user.

```
int countLesser = 0, countEqual = 0, countGreater = 0;
while (temp > 0) {
    int digit = temp % 10;
    if (digit < k)
        countLesser++;
    else if (digit == k)
        countEqual++;
    else
        countGreater++;
    temp /= 10;
}
printf("Number = %d, k = %d\n", num, k);
printf("Count_lesser = %d, Count_equal = %d, Count_greater = %d\n",
       countLesser, countEqual, countGreater);
break;
}
case 3:
    printf("Exiting program...\n");
    break;
default:
    printf("Invalid choice! Please enter 1, 2 or 3.\n");
}
} while (choice != 3);

return 0;
}
[EOF
[adilshakil@ADILs-MacBook-Air Downloads %
[adilshakil@ADILs-MacBook-Air Downloads % gcc menu_digit.c -o menu_digit
[adilshakil@ADILs-MacBook-Air Downloads %
```

Downloads — menu_digit — 120x30

```
        countEqual++;
    else
        countGreater++;
    temp /= 10;
}
printf("Number = %d, k = %d\n", num, k);
printf("Count_lesser = %d, Count_equal = %d, Count_greater = %d\n",
       countLesser, countEqual, countGreater);
break;
}
case 3:
    printf("Exiting program...\n");
    break;
default:
    printf("Invalid choice! Please enter 1, 2 or 3.\n");
}
} while (choice != 3);

return 0;
}
```

[EOF

[adilshakil@ADILs-MacBook-Air Downloads %

[adilshakil@ADILs-MacBook-Air Downloads % gcc menu_digit.c -o menu_digit

[adilshakil@ADILs-MacBook-Air Downloads % ./menu_digit

----- MENU -----

1. Sum and Product of first and last digit
2. Count digits lesser, equal, greater than k
3. Exit

Enter your choice: █

```
Downloads — -zsh — 120x30
1. Sum and Product of first and last digit
2. Count digits lesser, equal, greater than k
3. Exit
Enter your choice: Enter a positive integer: Number = 12345
First Digit = 1, Last Digit = 5
Sum = 6, Product = 5

[Restored 5 Sep 2026 at 8:33:31 PM]
Last login: Sat Sep  5 18:23:49 on ttys000
Restored session: Sat Sep  5 20:33:23 IST 2026
adilshakil@ADILs-MacBook-Air Downloads % ./menu_digit

----- MENU -----
1. Sum and Product of first and last digit
2. Count digits lesser, equal, greater than k
3. Exit
Enter your choice: 2
Enter a positive integer: 12345
Enter digit k (0-9) to compare: 3
Number = 12345, k = 3
Count_lesser = 2, Count_equal = 1, Count_greater = 2

----- MENU -----
1. Sum and Product of first and last digit
2. Count digits lesser, equal, greater than k
3. Exit
Enter your choice: 3
Exiting program...
adilshakil@ADILs-MacBook-Air Downloads %
```

Result

The program was compiled and executed successfully. The output for all test cases matched the expected output.